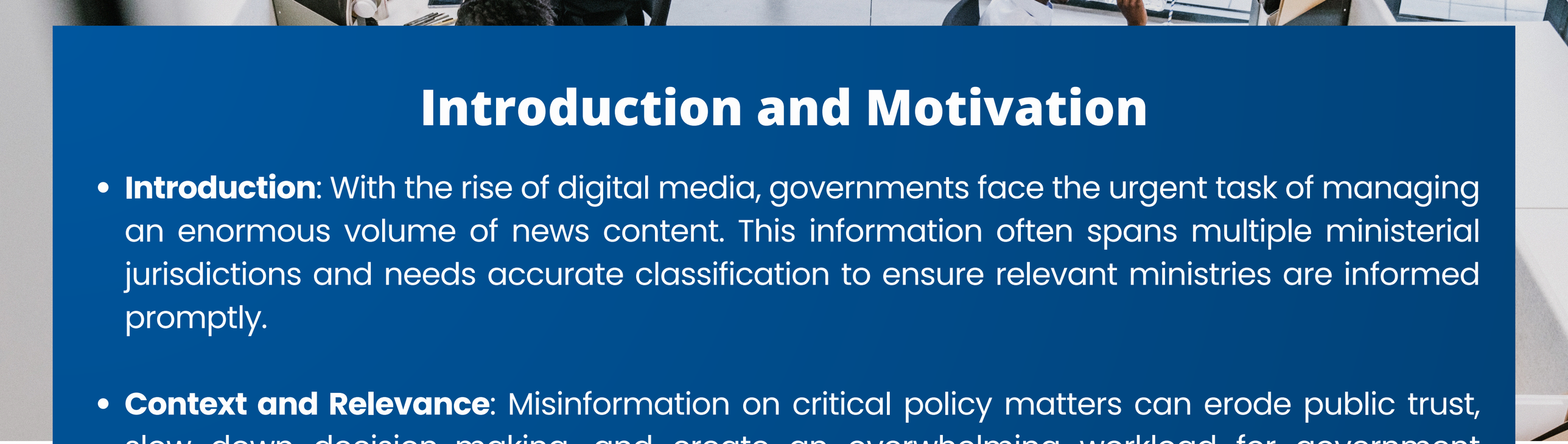
# DemocraticNet: A Deep Learning Pipeline for Multi-Class Multi-Label Ministry News Classification





# Introduction and Motivation

- **Introduction**: With the rise of digital media, governments face the urgent task of managing an enormous volume of news content. This information often spans multiple ministerial jurisdictions and needs accurate classification to ensure relevant ministries are informed promptly.

- **Context and Relevance**: Misinformation on critical policy matters can erode public trust, slow down decision-making, and create an overwhelming workload for government officials. Rapid, accurate news classification can support timely, coordinated responses to these issues.

- **Research Gap**: Existing studies on news classification focus on single-label categorization (like “real” vs. “fake” news) and are insufficient for handling complex multi-label classifications across governmental ministries. DemocraticNet addresses this unique gap by classifying news into multiple relevant ministries to support real-time governmental needs.

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# Research Problem and Objectives

**Research Problem**: How can we develop an automated classification system that can assign a news item to multiple relevant ministries accurately and efficiently, helping government officials to respond to critical issues more effectively?

**Objective 01**

Develop a deep learning pipeline leveraging advanced transformer architectures (BERT, DeBERTaV3, RoBERTa, and XLNet) for accurate classification.

**Objective 02**

Benchmark performance of different transformer models in multi-label classification and identify the optimal model-optimizer combinations.

**Objective 03**

Ensure high classification accuracy while minimizing computational costs to facilitate real-time deployment within governmental frameworks.

# Literature Review / Background

- **Overview of Related Work**: Previous studies in news classification have largely focused on simpler binary or single-label classifications, such as detecting misinformation on social media or categorizing news by general themes. These approaches fall short in addressing the complexity of classifying news into multiple ministerial domains.
- **Positioning DemocraticNet**: By adapting state-of-the-art transformer models specifically for multi-label classification tasks, DemocraticNet moves beyond previous studies to tackle the unique challenges of government news classification.
- **Novelty and Contribution**: DemocraticNet pioneers a system optimized for categorizing news articles relevant to government ministries. Unlike existing systems, it can handle cross-ministerial classifications in real-time, making it a novel and essential tool for governmental communication and misinformation management.

# Methodology

- **Approach Overview**: Our methodology involves fine-tuning multiple transformer models (BERT, DeBERTaV3, RoBERTa, XLNet) on a custom-built dataset of news articles tagged by ministerial relevance.
- **Data Collection and Preprocessing:**
  - Developed a dataset of 36,051 news articles, manually labeled for relevance to 59 ministries and an additional category for “no ministry” cases.
  - Conducted extensive data preprocessing to address class imbalances, removing underrepresented ministry classes to ensure fair and accurate model training.
- **Training Strategy**:
  - Fine-tuned models using a distributed GPU setup to reduce training time and increase efficiency.
  - Models were trained with various optimizers (SGD, Adam, AdamW, NAdam) to identify the combinations that provided the best performance.
- **Tools and Frameworks**: We used PyTorch, Hugging Face Transformers, and multi-GPU environments, leveraging NVIDIA RTX A5000 and T4 GPUs for parallel processing.

# Data Set

| Label | Ministry |
|---|---|
| 0 | Ministry of Railways |
| 1 | Ministry of Rural Development |
| 2 | Ministry of Steel |
| 3 | Ministry of Science & Technology |
| 4 | Ministry of Information & Broadcasting |
| 5 | Ministry of Food Processing Industries |
| 6 | Ministry of Health and Family Welfare |
| 7 | Ministry of Human Resource Development |
| 8 | Ministry of Agriculture |
| 9 | Ministry of Environment and Forests |
| 10 | Ministry of Chemicals and Fertilizers |
| 11 | Ministry of Water Resources |
| 12 | Ministry of Defence |
| 13 | Ministry of Petroleum & Natural Gas |
| 14 | President's Secretariat |
| 15 | Ministry of Micro, Small & Medium Enterprises |
| 16 | Ministry of Mines |
| 17 | Ministry of Tourism |
| 18 | Ministry of Housing & Urban Affairs |
| 19 | Ministry of Coal |
| 20 | Prime Minister's Office |
| 21 | Ministry of Textiles |
| 22 | Ministry of Commerce & Industry |
| 23 | Ministry of External Affairs |
| 24 | Ministry of Social Justice & Empowerment |
| 25 | Ministry of Power |
| 26 | Ministry of Consumer Affairs, Food & Public Distribution |
| 27 | Ministry of Heavy Industries & Public Enterprises |
| 28 | Ministry of Communications |
| 29 | Ministry of Shipping |
| 30 | Ministry of Finance |
| 31 | Ministry of Tribal Affairs |
| 32 | Ministry of Statistics & Programme Implementation |
| 33 | Ministry of Labour & Employment |
| 34 | Ministry of Law & Justice |
| 35 | Vice President's Secretariat |
| 36 | Ministry of Civil Aviation |
| 37 | Ministry for Development of North-East Region |
| 38 | UPSC Department |
| 39 | Ministry of Agro & Rural Industries |
| 40 | Ministry of Home Affairs |
| 41 | Ministry of Youth Affairs and Sports |
| 42 | Special Service and Features |
| 43 | Ministry of New and Renewable Energy |
| 44 | Ministry of Parliamentary Affairs |
| 45 | Planning Commission |
| 46 | Ministry of Personnel, Public Grievances & Pensions |
| 47 | Election Commission |
| 48 | Department of Space |
| 49 | Ministry of Disinvestment |
| 50 | Department of Ocean Development |
| 51 | Ministry of Overseas Indian Affairs |
| 52 | Ministry of Housing and Urban Poverty Alleviation |
| 53 | Ministry of Culture |
| 54 | Ministry of Company Affairs |
| 55 | Ministry of Panchayati Raj |
| 56 | Cabinet Committee on Economic Affairs (CCEA) |
| 57 | Cabinet |
| 58 | Department of Atomic Energy |
| 59 | Cabinet Committee Decisions |
| 60 | No Ministry |

Figure 1: Distribution of News Across Different Ministries (Classes), with 60 Indicating "No Ministry"

# Results and Analysis

**Table 3.** Experimental Results on the Newly Labeled News Dataset

| Model Name | Optimizer | Top-3 Accuracy | Loss |
|---|---|---|---|
| Deberta-base | SGD | 58.700% | 13.458% |
| Deberta-base | Adam | 82.809% | 6.593% |
| Deberta-base | Nadam | 82.683% | 6.685% |
| Deberta-base | AdamW | 78.229% | 9.752% |
| Roberta-base | SGD | 07.608% | 7.603% |
| Roberta-base | Adam | 81.915% | 7.077% |
| Roberta-base | NAdam | 81.273% | 7.878% |
| Roberta-base | AdamW | 82.544% | 6.551% |
| Xlnet-base-cased | SGD | 58.385% | 15.078% |
| Xlnet-base-cased | Adam | 79.974% | 9.806% |
| Xlnet-base-cased | Nadam | 84.583% | 6.282% |
| Xlnet-base-cased | AdamW | 79.318% | 9.945% |
| Bert-base-uncased | SGD | 29.842% | 76.227% |
| Bert-base-uncased | Adam | 77.810% | 9.672% |
| Bert-base-uncased | Nadam | 81.650% | 7.593% |
| Bert-base-uncased | AdamW | 79.220% | 9.334% |
| Xlm-roberta-base | SGD | 02.681% | 7.7984% |
| Xlm-roberta-base | Adam | 81.985% | 7.798% |
| Xlm-roberta-base | Nadam | 85.043% | 6.063% |
| Xlm-roberta-base | AdamW | 81.985% | 7.798% |

# Results and Analysis

- **Findings Summary**: Among the models tested, XLM-RoBERTa combined with the NAdam optimizer achieved the highest accuracy, with a top-3 accuracy score of 85.04% and a cross-entropy loss of 6.06. This model outperformed others in both accuracy and generalization.

- **Performance Comparison**:
  - DeBERTa and RoBERTa also performed well but showed slightly lower accuracy than XLM-RoBERTa.
  - Traditional optimizers like SGD were less effective, showing lower accuracy and higher loss compared to adaptive optimizers.

- **Significance**: These results confirm the feasibility of using transformer-based models for multi-class, multi-label news classification, showing that DemocraticNet can serve as a reliable tool for supporting government officials in managing and classifying news efficiently.

# Discussion

- **Interpreting Results**: Our results indicate that adaptive optimizers (e.g., Adam, NAdam) significantly improve model performance by effectively navigating complex loss landscapes. The success of XLM-RoBERTa underscores the model's ability to capture nuanced language patterns relevant to ministry classification.

- **Insights**: Key insights include the importance of balancing dataset class distribution and the performance gains achieved through specific model-optimizer combinations.

# Limitations and Future Work

- **Limitations**:
  - Dataset Constraints: Our dataset is limited to 36,051 articles, with an inherent class imbalance due to the underrepresentation of certain ministries.
  - Contextual Limitations: While effective, transformer models may struggle with context-specific interpretations, which could impact classification accuracy in certain cases.

- **Future Research Directions**:
  - Expand Dataset: Increase dataset size and diversity to improve robustness and model generalizability.
  - Comprehensive Ministry Coverage: Incorporate more ministries, covering a broader spectrum of news topics and improving overall government communication.
  - International Deployment: Adapt DemocraticNet's framework for use in other democratic contexts, allowing the model to serve governments worldwide.

# Conclusion

- **Summary**: DemocraticNet advances governmental news classification by automating the multi-label classification of news items across ministries, achieving high accuracy with minimal loss. This research contributes a scalable, adaptable solution to governmental information management.

- **Significance**: The project strengthens governmental transparency, supporting officials in managing news and misinformation more effectively. By applying deep learning to improve governmental workflows, DemocraticNet has the potential to bolster public trust in government communications.

- **Closing Statement**: DemocraticNet paves the way for automated, efficient, and transparent news classification in governmental settings, contributing a vital tool to support democratic values in an increasingly complex information landscape.

# Questions?

Thank you for your time and attention. I'm happy to answer any questions you may have about DemocraticNet.